

Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python
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Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham, United Kingdom
Durham University Sep 2025 – Present

- **Modules:** Bayesian ML (Foundations/Unsupervised; Regression/Classification), Optimisation & Control for AI, Deep Learning for Engineering, HPC/GPU Programming, Performance & Vectorisation.

MEng Electronic Engineering Durham, United Kingdom
Durham University Sep 2020 – Jun 2024

Experience

Industrial Tutor Durham, United Kingdom
Durham University, Department of Engineering Sep 2025 – Present

- Lead weekly design tutorials; mentor teams on scope, feasibility, Gantt planning, and design reviews.
- Coach teams through feasibility-driven decision-making: surface assumptions and specifications, test them, and retire designs that fail cost, risk, or performance constraints.

Graduate Communications Engineer Hebburn, United Kingdom
Siemens PLC Sep 2024 – Present

- Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools.
- Built Python-based engineering tooling to structure and query internal technical documentation and codebases using a retrieval-augmented workflow.
- Integrated local LLMs (via Ollama) with a vector database (Qdrant) for context-aware document retrieval, focusing on traceability and reproducible behaviour rather than model optimisation.
- Adapted existing network signal and waveform-generation utilities to interface with specific IP endpoints for development and testing workflows.
- Supported validation and testing activities by exercising communication paths and identifying failure modes during integration.

Projects & Research

Silicon Carbide JFET CPU Durham, United Kingdom
Master's Dissertation Oct 2023 – Apr 2024

- Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments.
- Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction.
- Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Scientific Computing; Python; C++; High-Performance Computing

Leadership, Activities & Interests

Leadership: Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).

Activities: Durham University Fencing Team; Boxing (Student Fight Night).

Achievements: Weldon le Huray Fencing Scholarship (2020–24); South Africa U17 Fencing Champion; President's Award (Gold).